
Experiment No:- 4

Aim / Title:- To use function types, scope, and closures; apply try-catch; build a palindrome checker.

Software / Tools Required

- VS Code (Live Server Extension)
- Google Chrome / Mozilla Firefox
- HTML5, CSS3, JavaScript (ES6+) | OS: Windows 10/11

Theory:- A palindrome is a word or number that reads the same forwards and backwards, ignoring case and spaces. This experiment demonstrates four core JavaScript concepts:

- 1. Function Types:** Named functions are hoisted to the top of scope. Arrow functions `() => {}` offer concise syntax with lexical this binding.
- 2. Scope:** Variables declared with `let` and `const` are block-scoped — they cannot be accessed outside their declared block, preventing global pollution.
- 3. Closures:** A closure is formed when an inner function retains access to its outer function's variables after the outer function has returned. The check counter persists across calls via closure.
- 4. Exception Handling (try-catch):** Runtime errors such as passing invalid types are caught gracefully using `try-catch`, displaying a friendly error message without crashing the app.

Output

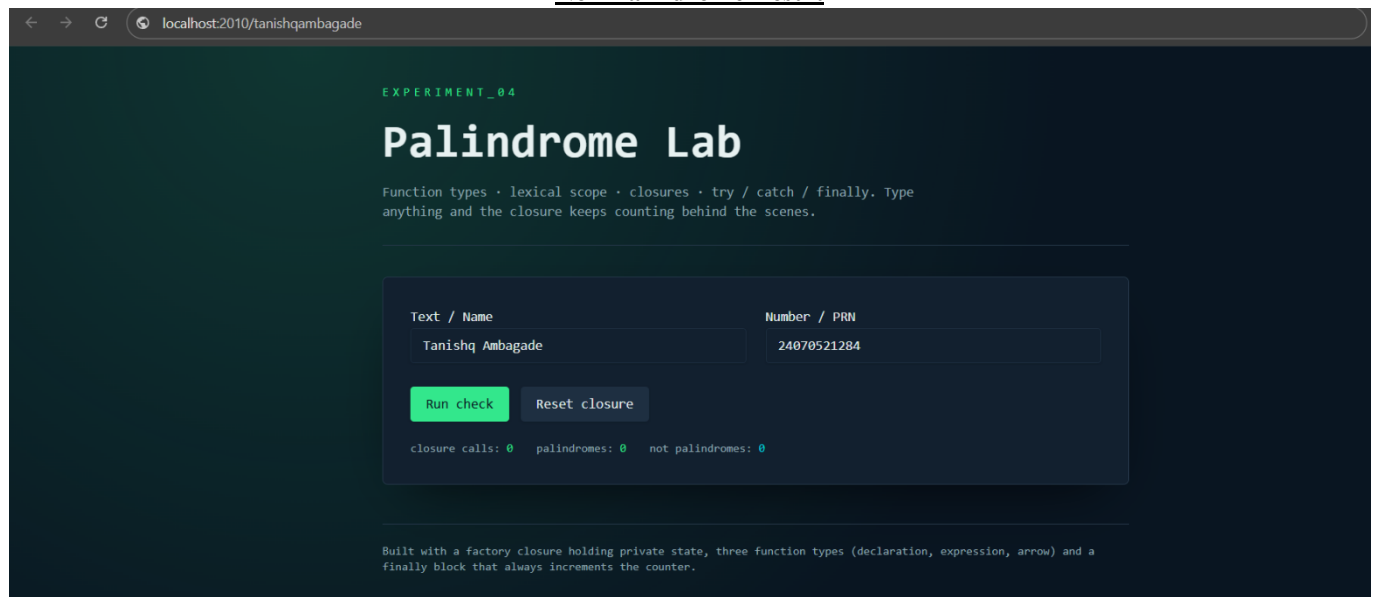
File Path: file:///C:/Users/ASUS/jsex4/index.html/tanishqambagade

Test 1: "Tanishq Ambagade" / "24070521284" → Both are not Palindromes (NO MATCH)

Test 2: "Noon" / "1234321" → Both are Palindromes (MATCH)

Outputs:-

Non-Palindrome Result



← → ↻ 🌐 localhost:2010/tanishqambagade

Run check

Reset closure

closure calls: 2 palindromes: 0 not palindromes: 2

NAME • CALL #1

NOT A PALINDROME

input	Tanishq Ambagade
normalized	tanishqambagade
reversed	edagabmaqhsinat
length	15

PRN • CALL #2

NOT A PALINDROME

input	24070521284
normalized	24070521284
reversed	48212507042
length	11

Palindrome Result

← → ↻ 🌐 localhost:2010/tanishqambagade

Text / Name

Number / PRN

Noon

1234321

Run check

Reset closure

closure calls: 4 palindromes: 2 not palindromes: 0

NAME • CALL #3

PALINDROME

input	Noon
normalized	noon
reversed	noon
length	4

PRN • CALL #4

PALINDROME

input	1234321
normalized	1234321
reversed	1234321
length	7

Edit w

Additional Assignment:- Study VS Instagram Calulator

[←](#) [→](#) [↻](#) localhost:1020/tanishqambagade

Study vs Instagram

Where does your time really go?

Enter your daily study and Instagram hours. We'll tell you whether you deserve a break or need to hit the books.

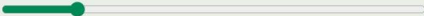


[←](#) [→](#) [↻](#) localhost:1020/tanishqambagade

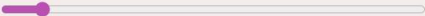
Study vs Instagram



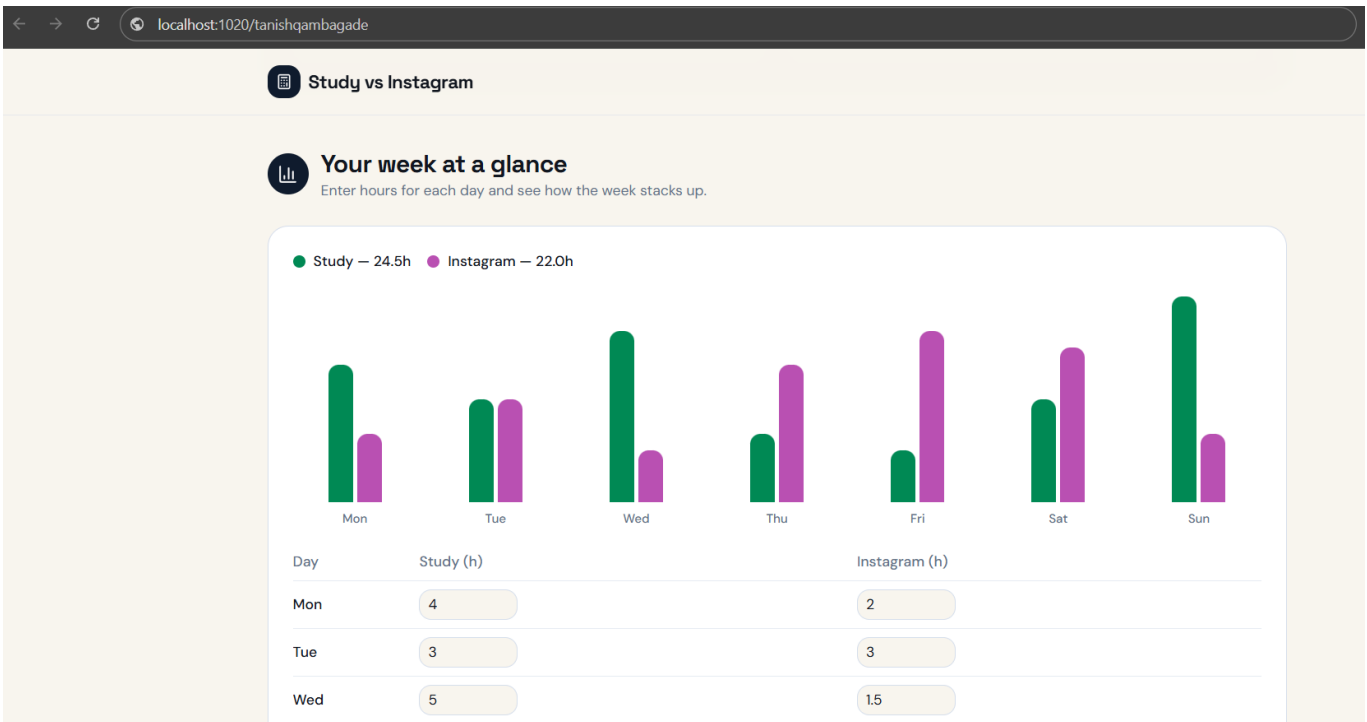
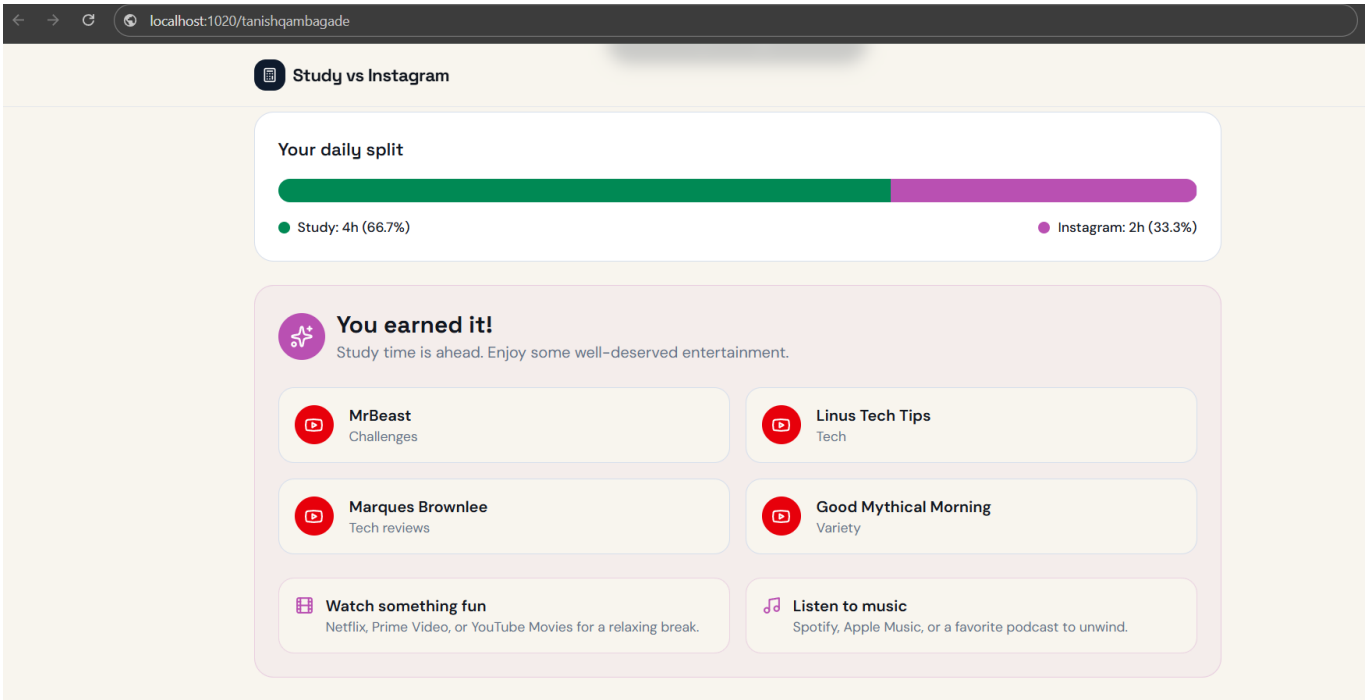
**Study Time**
Hours spent learning today

0h  24h

**Instagram Time**
Hours spent scrolling today

0h  24h

 **Compare my time** →



Study vs Instagram

Thu	2	4
Fri	1.5	5
Sat	3	4.5
Sun	6	2

Weekly rating: Good
Study time is ahead by 2.5h this week. Solid balance.

Social app time log

Log when you opened and closed each app to see exactly how much time it took.

App

Instagram

Started

09:15

Ended

10:05

50m

App

WhatsApp

Started

13:00

Ended

13:25

25m

Study vs Instagram

App

Instagram

Started

09:15

Ended

10:05

50m

App

WhatsApp

Started

13:00

Ended

13:25

25m

App

Facebook

Started

21:30

Ended

22:10

40m

+ Add session

Total social media time: 1h 55m

Instagram

50m

WhatsApp

25m

Facebook

40m

Use this as today's social time

Result / Conclusion:- The webpage was successfully created and executed. The output verifies the correct implementation of JavaScript function types, variable scope, and closures. The experiment also demonstrates effective exception handling using try-catch and successfully validates user input to determine whether the entered string is a palindrome, showcasing the practical application of modern JavaScript programming concepts.